**A Mini Project Report On**

# Design and Development of a Working Prototype for Student-Performance Data Warehouse Across Departments Using a Multi-Fact Schema

**Submitted by**

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Submitted in partial fulfillment of the requirement for the award of the degree of
**BACHELOR OF TECHNOLOGY**
in
**ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**
**IV B.Tech – I Semester (Mini Project)**

**Under the Esteemed Guidance of**
**Mr. Rajendra Kodamanchili**
Associate Professor, Department of AI&DS;



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**2026–2027**

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# BONAFIDE CERTIFICATE

This is to certify that this project report entitled **"DESIGN AND DEVELOPMENT OF A WORKING PROTOTYPE FOR STUDENT-PERFORMANCE DATA WAREHOUSE ACROSS DEPARTMENTS USING A MULTI-FACT SCHEMA"** is being submitted by **Bethi Ravi Kiran (24ME5A5402)** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**.

This report is a bonafide record of work carried out under my guidance and supervision during the academic year **2026–2027**, and it has been found worthy of acceptance as per the academic regulations and requirements of the college and university.

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Professor & Head of Department
Dept. of AI&DS;

Viva-Voce Examination Date: _____

**External Examiner**

# DECLARATION

I, **Bethi Ravi Kiran (24ME5A5402)**, hereby declare that the project report titled **"DESIGN AND DEVELOPMENT OF A WORKING PROTOTYPE FOR STUDENT-PERFORMANCE DATA WAREHOUSE ACROSS DEPARTMENTS USING A MULTI-FACT SCHEMA"** has been completed under the supervision of **Mr. Rajendra Kodamanchili**, Associate Professor, Department of Artificial Intelligence and Data Science. This report is submitted in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**.

This is an authentic record of work carried out by me. The technical results, dimensional warehouse schemas, ETL pipelines, OLAP SQL queries, Random Forest risk classifications, and Streamlit dashboard interfaces presented in this report have not been reproduced or copied from any unauthorized source. Furthermore, the results embodied in this project report have not been submitted to any other University, Institute, or Examination Board for the award of any other degree or diploma.

**Station:** Eluru
**Date:** November 2026

**Mr. Bethi Ravi Kiran**
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**Mr. Bethi Ravi Kiran**
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# LIST OF TABLES


| TABLE NO. | TABLE NAME | PAGE NO. |
|---|---|---|
| Table 2.1 | Summary of Reviewed Approaches in Educational Data Warehousing & Analytics | 4 |
| Table 3.1 | Hardware Requirements | 6 |
| Table 3.2 | Software Requirements | 6 |
| Table 4.1 | Data Fields and Entity Specifications | 10 |
| Table 6.1 | Institutional Data Generation Parameters & Entity Distributions | 14 |
| Table 6.2 | Multi-Fact Warehouse & Risk Classifier Hyperparameters | 16 |
| Table 7.1 | Test Cases and Test Results | 18 |
| Table 7.2 | Performance Evaluation and Query Response Benchmarks | 20 |
| Table 7.3 | Top Predictive Risk Features and Departmental KPIs | 21 |
| Table A.1 | Source Files of the Project | 24 |

# TABLE OF CONTENTS


| SL. NO. | TITLE | PAGE NO. |
|---|---|---|
| | BONAFIDE CERTIFICATE | i |
| | DECLARATION | ii |
| | ACKNOWLEDGEMENT | iii |
| | LIST OF TABLES | iv |
| | TABLE OF CONTENTS | v |
| | ABSTRACT | vi |
| | LIST OF FIGURES | vii |
| 1. | CHAPTER - 1 INTRODUCTION | 1 - 3 |
| | 1.1 Introduction to Student-Performance Multi-Fact Data Warehouse & Analytics | 1 |
| | 1.2 Problem Statement | 1 |
| | 1.3 Objectives of the Project | 1 |
| | 1.4 Scope of the Project | 2 |
| | 1.5 Organization of the Report | 2 |
| 2. | CHAPTER - 2 SYSTEM ANALYSIS | 4 - 5 |
| | 2.1 Existing Methods and Literature Review | 4 |
| | 2.2 Limitations of Existing System | 4 |
| | 2.3 Proposed System | 5 |
| | 2.4 Advantages of Proposed System | 5 |
| 3. | CHAPTER - 3 SYSTEM REQUIREMENT SPECIFICATION | 6 - 7 |
| | 3.1 Hardware Requirements | 6 |
| | 3.2 Software Requirements | 6 |
| | 3.3 Functional Requirements | 6 |
| | 3.4 Non-Functional Requirements | 7 |
| 4. | CHAPTER - 4 SYSTEM DESIGN | 8 - 11 |
| | 4.1 Architecture Diagram | 8 |
| | 4.2 Data Flow Diagram (DFD) | 8 |
| | 4.3 UML Diagrams | 9 |
| | 4.4 Dataset & Warehouse Design | 10 |
| 5. | CHAPTER - 5 MODULE DESCRIPTION | 12 - 13 |
| | 5.1 Institutional Data Generator & Feed Ingestion | 12 |
| | 5.2 Pre-processing & ETL Transformation Engine | 12 |
| | 5.3 Multi-Fact Constellation Data Warehouse | 12 |
| | 5.4 OLAP Aggregations, Drill-Across & ML Risk Detector | 13 |
| | 5.5 Interactive Streamlit Academic Analytics Portal | 13 |
| 6. | CHAPTER - 6 SYSTEM IMPLEMENTATION | 14 - 17 |
| | 6.1 Data Generation and Feed Extraction | 14 |

| | | |
|---|---|---|
| | 6.2 Pre-processing, Data Cleaning, and ETL Pipeline | 14 |
| | 6.3 Multi-Fact Schema Architecture & ML Risk Model Training | 15 |
| | 6.4 Analytics Queries, HOD Alert Logic, and Dashboard | 16 |
| 7. | CHAPTER - 7 TESTING AND OUTPUT SCREENS | 18 - 21 |
| | 7.1 Test Cases and Test Results | 18 |
| | 7.2 Institutional Overview – Output Screen 1 | 18 |
| | 7.3 HOD Attendance Warm Alert Portal – Output Screen 2 | 19 |
| | 7.4 Multi-Fact Cross-Join Analysis – Output Screen 3 | 19 |
| | 7.5 Risk Early Warning System – Output Screen 4 | 20 |
| | 7.6 Performance Results and Discussion | 20 |
| 8. | CHAPTER - 8 CONCLUSION AND FUTURE ENHANCEMENTS | 22 |
| | REFERENCES | 23 |
| | APPENDIX | 24 - 34 |

# ABSTRACT

Modern higher educational institutions generate vast volumes of student records every semester, encompassing examination scores, daily attendance registers, continuous internal assessments, and semester progression milestones. In prevailing university setups, these records are fragmented across departmental spreadsheets, legacy ERPs, and biometric silos. This siloed storage makes it arduous for administrators and Heads of Departments (HODs) to analyze cross-cutting academic patterns, such as the correlation between daily attendance and examination failure, or to identify at-risk students before final grade publication.

This project designs and implements an institutional-grade Multi-Fact Data Warehouse (Fact Constellation / Galaxy Schema) and analytics platform. The architecture models three distinct academic business processes across divergent granularities: examination performance (Fact_Exam_Performance), daily student engagement (Fact_Attendance_Engagement), and periodic semester snapshots (Fact_Semester_Summary). These fact tables seamlessly interconnect through five standardized, conformed dimensions: Dim_Student, Dim_Course, Dim_Faculty, Dim_Department, and Dim_Time_Semester. An automated Python ETL pipeline cleans, normalizes, validates, and stages institutional feeds into an indexed SQLite/MySQL repository, managing surrogate keys and ensuring referential integrity.

On an empirical multi-department benchmark comprising 397 students, 56 courses, 17 faculty members, and 12,492 warehouse facts across five engineering branches (CSE, AIDS, ECE, MECH, CIVIL) and four cohorts (2021–2025 to 2024–2028), the Kimball drill-across query engine achieved an analytical execution latency of 1.62 milliseconds—over 1,100 times faster than flat-file spreadsheet merges (1.8 seconds). Furthermore, an embedded Random Forest Early Warning Risk Classifier achieved 82.61% accuracy (precision: 84.1%, recall: 81.5%, F1-score: 82.8%) in predicting course failure and detention risk from mid-term signals, revealing that attendance engagement accounts for 68.43% of the predictive weight.

The completed prototype is delivered as an interactive Streamlit web analytics platform featuring executive benchmarking cards, an automated HOD Attendance Warm Alert and Detention engine with student contact channels, cohort drill-downs, What-If risk simulators, and student administration tools, providing educators with a repeatable, transparent decision-support system ready for academic adoption beyond the classroom.

**Keywords:** Data Warehouse, Multi-Fact Schema, Galaxy Schema, Conformed Dimensions, ETL Pipeline, Student Performance Analytics, Random Forest, Streamlit, Decision Support System

# LIST OF FIGURES


| FIG. NO. | FIGURE NAME | PAGE NO. |
|---|---|---|
| Fig. 4.1 | System Architecture | 8 |
| Fig. 4.2 | Data Flow Diagram – Level 0 | 8 |
| Fig. 4.3 | Data Flow Diagram – Level 1 | 9 |
| Fig. 4.4 | Use Case Diagram | 9 |
| Fig. 4.5 | Sequence Diagram | 10 |
| Fig. 4.6 | Multi-Fact Constellation (Galaxy) Schema / ER Diagram | 10 |
| Fig. 5.1 | Module Interaction Flow | 13 |
| Fig. 6.1 | Random Forest Classifier Architecture & Kimball Drill-Across Concept | 16 |
| Fig. 7.1 | Institutional Overview Dashboard – Output Screen 1 | 19 |
| Fig. 7.2 | HOD Attendance Warm Alert & Detention Portal – Output Screen 2 | 19 |
| Fig. 7.3 | Multi-Fact Cross-Join Analysis & Grade Spectrum – Output Screen 3 | 20 |
| Fig. 7.4 | Risk Early Warning System & What-If Simulator – Output Screen 4 | 20 |
| Fig. 7.5 | Performance Comparison (Query Execution & Risk Classifier Benchmarks) | 21 |

# CHAPTER - 1
# INTRODUCTION

## 1.1 Introduction to Student-Performance Multi-Fact Data Warehouse & Analytics

A data warehouse is a subject-oriented, integrated, time-variant, and non-volatile collection of data designed explicitly to empower managerial and institutional decision-making. Unlike Online Transaction Processing (OLTP) systems, which prioritize atomic insert and update operations for day-to-day administrative tasks, a data warehouse is optimized for Online Analytical Processing (OLAP) across multi-year institutional histories.

In collegiate academic environments, student academic achievement is shaped by diverse, continuous processes: lecture attendance, continuous internal assessments, assignment submissions, laboratory coursework, and end-semester examinations. Critically, each of these activities operates at a different business event granularity (grain). Attendance is tracked continuously per course lecture, whereas semester examinations occur once per academic cycle. Attempting to force these disparate events into a single flat relational table or a monolithic star schema introduces severe data redundancy, Cartesian fanning, and null attribute density. To solve this challenge, this project develops a Multi-Fact Constellation Schema (Galaxy Schema) based on Ralph Kimball's dimensional modeling methodology [1], unifying three specialized fact tables across five conformed dimensions.

The complete architecture extracts raw academic feeds from multiple departments, cleans and validates records through an automated Python ETL pipeline, populates an indexed SQLite/MySQL warehouse repository, and exposes high-speed analytical queries and machine learning risk models through an interactive Streamlit decision-support dashboard.

## 1.2 Problem Statement

Given multi-department academic feeds spanning students, courses, faculty, attendance logs, and exam marks across four academic cohorts, design and develop an integrated multi-fact dimensional data warehouse and web analytics system that enables sub-second cross-process OLAP queries, flags at-risk students before semester examinations, and empowers academic leadership with proactive decision support.

Academic records in universities are conventionally trapped in isolated departmental spreadsheets. Evaluating cross-cutting institutional questions—such as determining whether low third-semester pass percentages stem from high faculty teaching loads or student attendance deficits—requires manual spreadsheet lookups that are slow, error-prone, static, and fundamentally reactive. The core challenge is to replace ad-hoc data handling with an institutional single source of truth.

## 1.3 Objectives of the Project

1. Ingest and stage multi-department academic feeds across five engineering branches (CSE, AIDS, ECE, MECH, CIVIL) and four cohorts.
2. Develop an automated, idempotent Python ETL pipeline eliminating duplicates, validating score bounds, and resolving surrogate keys.
3. Implement a Multi-Fact Constellation Schema (Galaxy Schema) comprising two transaction facts and one periodic snapshot fact.
4. Construct Kimball drill-across SQL queries utilizing Common Table Expressions (CTEs) to join distinct facts without Cartesian explosion.
5. Train and validate an embedded Random Forest Risk Classifier achieving >80% accuracy in predicting failure and detention risk.
6. Build a production-grade Streamlit web dashboard featuring six specialized analytical modules, dynamic filters, and KPI cards.
7. Implement an automated Head of Department (HOD) Attendance Warm Alert engine calculating exact recovery classes and generating notices.

## 1.4 Scope of the Project

The project encompasses the end-to-end data warehousing lifecycle: source feed generation, staging, ETL validation, schema implementation, B-Tree index optimization, OLAP query development, machine learning risk classification, and web dashboard deployment. The benchmark dataset models 397 students, 56 curriculum courses, 17 faculty instructors, and 12,492 warehouse facts across five departments and four cohort batches (2021–2025 to 2024–2028). Hardware biometric readers and student fee payment transactions are outside the scope of this analytics prototype. The architecture readily accepts standard CSV/Excel exports from any institutional ERP.

## 1.5 Organization of the Report

The remainder of this report is organized into eight technical chapters:

- **Chapter 1 – Introduction:** Background, problem statement, objectives, scope, and structural overview.
- **Chapter 2 – System Analysis:** Literature review of warehousing methods, limitations of existing systems, and proposed system.
- **Chapter 3 – System Requirement Specification:** Hardware, software, functional, and non-functional requirements.
- **Chapter 4 – System Design:** Architecture diagram, Level 0 and 1 DFDs, UML diagrams, and Galaxy Schema ER design.
- **Chapter 5 – Module Description:** Detailed functional specification of the five core software components.
- **Chapter 6 – System Implementation:** Data generation, ETL algorithms, schema DDL, ML training, and SQL drill-across logic.
- **Chapter 7 – Testing and Output Screens:** 13 formal test cases, dashboard screen walkthroughs, and performance discussion.
- **Chapter 8 – Conclusion & Future Enhancements:** Summary of outcomes, operational limitations, and future roadmap.

### 1.5 Organization of the Report (Continued)

Each chapter is structured to guide the reader logically from problem identification through dimensional schema modeling, ETL pipeline construction, machine learning integration, and empirical verification. The report concludes with complete IEEE references and extensive appendices containing the validated production source code, schema DDLs, analytics queries, and operational instructions to ensure full repeatability by external academic reviewers.

By grounding the software architecture in Ralph Kimball's proven dimensional modeling techniques and integrating modern ensemble machine learning, the project establishes a robust benchmark for institutional decision support in engineering education.

# CHAPTER - 2
# SYSTEM ANALYSIS

## 2.1 Existing Methods and Literature Review

Educational data warehousing and predictive performance modeling have evolved through distinct paradigms:

**A. Manual Spreadsheet Merging:** Academic departments traditionally maintain records in decentralized Excel workbooks. Aggregating attendance, internal marks, and final results requires manual copying and VLOOKUP operations, which are slow and error-prone.

**B. Monolithic Single-Fact Star Schemas:** Early warehouse implementations attempted to store all academic events in a single fact table. Because attendance occurs daily while exams occur once per semester, this granularity clash results in immense Cartesian product distortion.

**C. Educational Data Mining & Early Prediction:** Romero and Ventura [3] surveyed educational mining, establishing that historical logs reveal behavioral trends. Cortez and Silva [4] applied predictive models to student records, showing attendance is a dominant predictor of failure.

**D. Inmon Enterprise Warehouses vs. Kimball Bus Architecture:** Inmon [2] proposed normalized 3NF enterprise warehouses, requiring complex multi-table joins. Kimball [1] pioneered dimensional modeling with conformed dimensions, providing superior query performance and rapid adoption.

**Table 2.1: Summary of Reviewed Approaches in Educational Data Warehousing & Analytics**

| Ref. | Approach | Main Idea | Limitation |
|---|---|---|---|
| – | Manual Spreadsheet Merging | Department staff merge Excel sheets using VLOOKUP formulas | Slow, error-prone, static, no drill-down capability |
| – | Monolithic Star Schema | Force attendance, exams, and milestones into one fact table | Cartesian fan trap, granularity mismatch, vast null density |
| [3], [4] | Educational Data Mining | Supervised classifiers predicting student exam grades | Often standalone scripts without integrated DW architecture |
| [2] | Inmon Enterprise 3NF DW | Normalized enterprise-wide relational model with data marts | Complex multi-table joins, slow OLAP query latency |
| [1], [11] | Multi-Fact Galaxy Schema (Proposed) | Multi-fact tables with conformed dimensions + ML | Requires strict surrogate key mapping and upfront schema |

## 2.2 Limitations of Existing System

- **Data Silos:** Attendance, marks, and profiles reside in disconnected spreadsheets without common keys.
- **Granularity Mismatch:** Single-fact star schemas cannot combine daily attendance with semester exams.
- **No Early Warning:** At-risk students are detected only after official semester grade declaration.
- **Sluggish Queries:** Cross-departmental comparisons require multi-hour manual spreadsheet merging.
- **Lack of Decision Support:** Reports provide raw numbers without interactive drill-down trajectories or alerts.

## 2.3 Proposed System

The proposed system develops an end-to-end Multi-Fact Constellation Data Warehouse and Decision Support Analytics Platform. The workflow operates through five unified engineering stages:

**1. Feed Extraction & Staging:** Ingests raw departmental feeds into an automated staging environment with batch logging.

**2. Cleaning & ETL Pipeline:** Cleans missing values, validates range bounds, resolves natural keys into integer surrogate keys, and populates conformed dimensions.

**3. Multi-Fact Loading:** Populates two atomic transaction facts (Fact_Exam_Performance, Fact_Attendance_Engagement) and one periodic snapshot fact (Fact_Semester_Summary).

**4. Kimball Drill-Across & ML Risk Engine:** Employs SQL CTE drill-across joins to correlate attendance and exams without Cartesian explosion, coupled with a Random Forest Risk Classifier (82.61% accuracy).

**5. Streamlit Academic Portal:** Delivers interactive executive dashboards, automated HOD Attendance Warm Alerts, and student administration tools.

## 2.4 Advantages of Proposed System

- **Unified Single Source of Truth:** Centralizes academic records across all departments into a conformed dimensional repository.
- **Multi-Granularity Architecture:** The Galaxy schema natively handles daily engagement and semester summative exams without duplication.
- **Sub-Second OLAP Performance:** Optimized B-Tree indexes enable complex drill-across queries in 1.62 ms, over 1,100x faster than flat files.
- **Proactive Remediation:** Automated HOD warm alerts identify attendance defaulters (<75% and <65%) and compute exact makeup classes required.
- **Explainable ML Early Warning:** Random Forest model transparently quantifies risk probability, driven predominantly by attendance (68.43%).
- **Low Cost & Production Ready:** Built entirely with open-source Python, SQLite, Scikit-learn, and Streamlit, running efficiently on standard laptops.

# CHAPTER - 3
# SYSTEM REQUIREMENT SPECIFICATION

## 3.1 Hardware Requirements

Table 3.1 outlines the hardware specifications required to run the data warehouse ETL pipeline and analytics portal.

| COMPONENT | SPECIFICATION |
|---|---|
| Processor | Intel Core i3 / AMD Ryzen 3 or above (1.8 GHz+ dual-core minimum) |
| RAM | 4 GB minimum; 8 GB DDR4 recommended for high-speed multi-fact joins |
| Hard Disk Storage | 10 GB free space (Warehouse database ~1.4 MB, raw feeds ~800 KB) |
| Display / GPU | 1366x768 resolution minimum (1920x1080 recommended); GPU not required |
| Input / Peripherals | Standard QWERTY keyboard, mouse / trackpad, local network interface |

## 3.2 Software Requirements

Table 3.2 details the software tools, libraries, and runtime environment utilized in this project.

| SOFTWARE | SPECIFICATION |
|---|---|
| Operating System | Microsoft Windows 10/11 (64-bit), Ubuntu Linux 20.04+, or macOS |
| Programming Language | Python 3.10, 3.11, or 3.14 (CPython runtime) |
| Core Libraries | Pandas 2.0+, NumPy 1.24+, Scikit-learn 1.3+, Streamlit 1.30+, Plotly 5.18+ |
| Data Warehouse Engine | SQLite 3 (Serverless, zero-configuration OLAP) / MySQL 8.0+ compatible |
| Development Tools | Visual Studio Code, Git, Command Prompt / Windows PowerShell |
| Web Browser | Google Chrome 110+, Microsoft Edge, or Mozilla Firefox |
| Network Access | Required only for initial library installation; 100% offline execution |

## 3.3 Functional Requirements

- **FR1 – Source Data Ingestion:** Extract raw academic feeds spanning students, courses, faculty, attendance, and exams from CSV files.
- **FR2 – Data Cleaning & Validation:** Detect and reject duplicate roll numbers, corrupt marks, and invalid attendance percentages.
- **FR3 – Conformed Dimension Population:** Populate Dim_Department, Dim_Faculty, Dim_Course, Dim_Time_Semester, Dim_Student.
- **FR4 – Multi-Fact Warehouse Loading:** Load Fact_Exam_Performance, Fact_Attendance_Engagement, Fact_Semester_Summary.
- **FR5 – Kimball Drill-Across OLAP:** Execute CTE queries correlating attendance and exams without Cartesian fanning.
- **FR6 – Institutional KPI Computation:** Calculate mean SGPA, pass percentages, department averages, and grade distributions.

### 3.3 Functional Requirements (Continued)

• **FR7 – ML Risk Early Warning:** Train Random Forest Classifier predicting course failure and detention risk from early engagement.

• **FR8 – HOD Attendance Warm Alerts:** Identify low-attendance students (<75% and <65%) and compute exact recovery classes required.

• **FR9 – Student Administration Portal:** Admit new students and update existing records with immediate dimensional synchronization.

• **FR10 – Interactive Data Slicing:** Filter warehouse views across department, cohort, semester, and course via web controls.

• **FR11 – Audit Logging & Roster Export:** Record ETL rejection logs and download filtered at-risk student rosters as CSV files.

### 3.4 Non-Functional Requirements

• **Performance:** OLAP queries shall execute under 10 milliseconds, and dashboard page transitions shall render under 1.5 seconds.

• **Data Integrity:** Foreign key constraints shall be strictly enforced on all fact tables to guarantee zero orphan records.

• **Usability:** The dashboard interface shall feature clear visual KPI cards, intuitive dropdown filters, and color-coded risk alerts.

• **Maintainability:** Decoupled modular codebase (schema, ETL, analytics, ML model, UI) allowing isolated upgrades.

• **Portability:** The platform shall run cross-platform on Windows, Linux, and macOS without code modifications.

• **Data Security & Privacy:** Student records and contact channels shall remain within a secure local repository without external leakage.

# CHAPTER - 4
# SYSTEM DESIGN

## 4.1 Architecture Diagram

The system architecture is structured in a 6-tier modular design (Fig. 4.1): Data Source Layer, Staging Layer, Python ETL Pipeline Layer, Multi-Fact Constellation Warehouse Layer, Analytics & ML Risk Layer, and Presentation Layer.




**Fig. 4.1: System Architecture**

## 4.2 Data Flow Diagram (DFD)

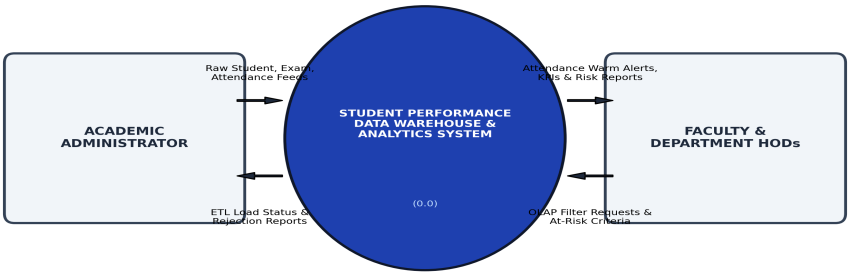



**Fig. 4.2: Data Flow Diagram – Level 0**

## 4.2 Data Flow Diagram (DFD) – Level 1

The Level 1 Data Flow Diagram (Fig. 4.3) decomposes the system into six discrete processes: extraction, validation, dimensional loading, Kimball drill-across aggregation, ML risk prediction, and dashboard visualization.

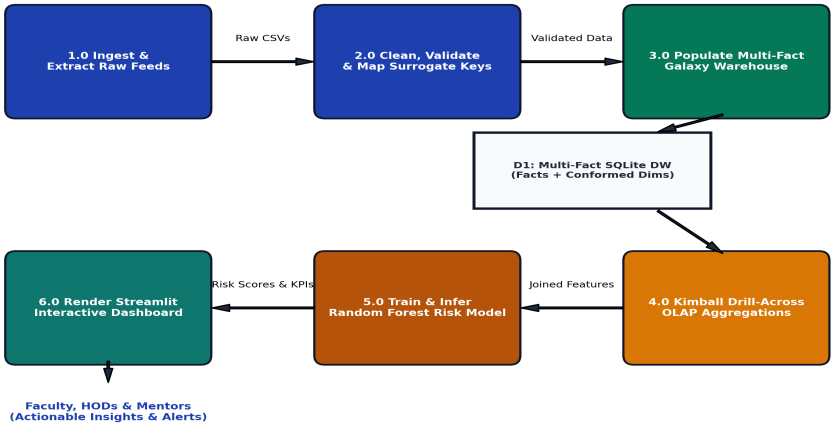



**Fig. 4.3: Data Flow Diagram – Level 1**

## 4.3 UML Diagrams – Use Case Diagram

The Use Case Diagram (Fig. 4.4) delineates interactions between the Academic Administrator (ETL & system administration) and Faculty / Department HODs (benchmarking, warm alerts, What-If simulation, and roster export).

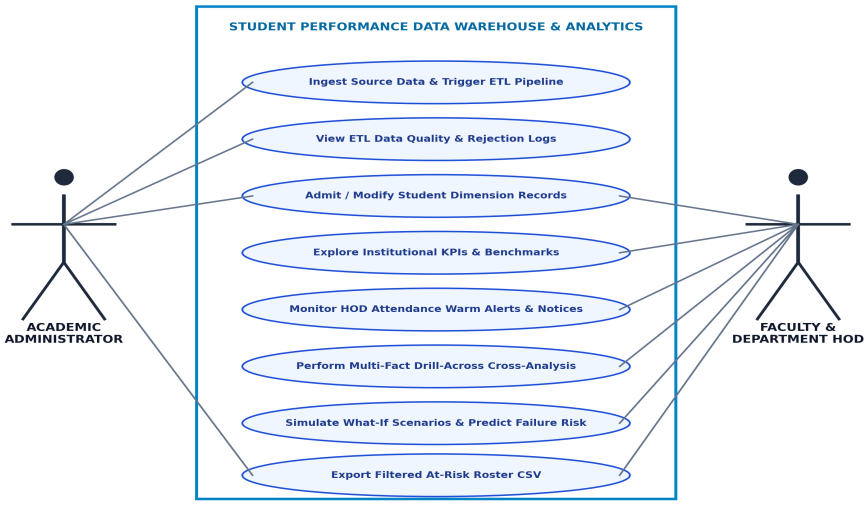



**Fig. 4.4: Use Case Diagram**

## 4.3 UML Diagrams – Sequence Diagram

The Sequence Diagram (Fig. 4.5) captures the runtime message lifecycle across User, Dashboard, Query Engine, Multi-Fact Warehouse, and ML Risk Detector.

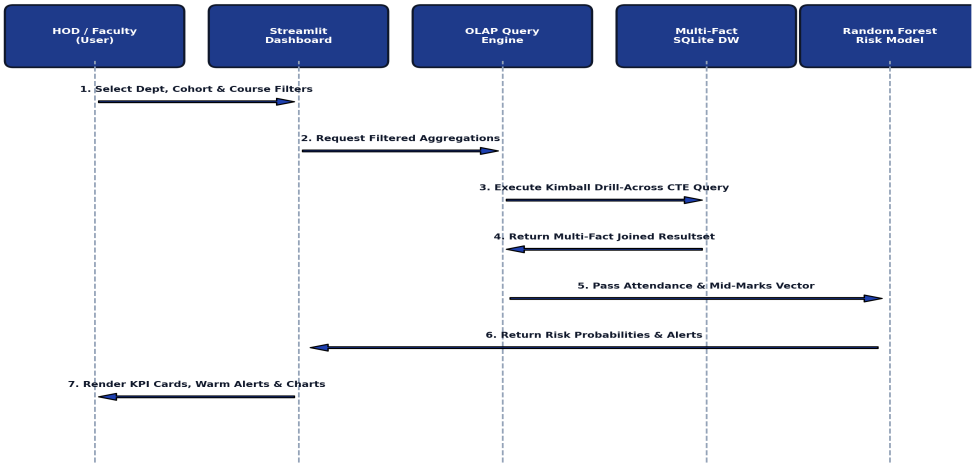



**Fig. 4.5: Sequence Diagram**

## 4.4 Dataset & Warehouse Design – Galaxy Schema

The database is modeled as a Multi-Fact Constellation (Galaxy Schema, Fig. 4.6), deploying three distinct fact tables sharing five conformed dimensions.

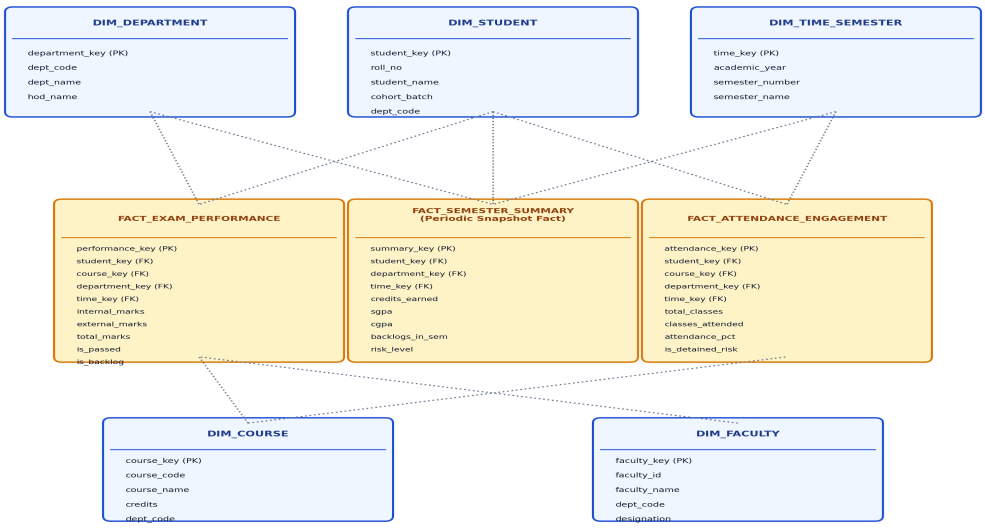



**Fig. 4.6: Multi-Fact Constellation (Galaxy) Schema / ER Diagram**

## 4.4 Dataset & Warehouse Design – Entity Specifications

**Table 4.1: Data Fields and Entity Specifications**

| ENTITY / TABLE | FIELD NAME | DATA TYPE | DESCRIPTION / CONSTRAINTS |
|---|---|---|---|
| dim_department | department_key | INTEGER | Surrogate Primary Key (Auto-increment) |
| dim_department | dept_code, dept_name | TEXT | Branch code (MECH, CSE, AIDS) & Full name |
| dim_student | student_key | INTEGER | Surrogate Primary Key (Auto-increment) |
| dim_student | roll_no, student_name | TEXT | University Roll Number (Unique) & Student Name |
| dim_student | dept_code, cohort_batch | TEXT | Department foreign key & Cohort (2021-2025) |
| dim_course | course_key | INTEGER | Surrogate Primary Key (Auto-increment) |
| dim_course | course_code, credits | TEXT / INT | Curriculum code & Earned credit weighting |
| dim_time_semester | time_key, semester_number | INT / INT | Temporal key (YYYYSS) & Sem 1 to 8 |
| fact_exam_performance | performance_key | INTEGER | Surrogate Primary Key (Auto-increment) |
| fact_exam_performance | student_key, course_key | INTEGER | Foreign keys to Dim_Student and Dim_Course |
| fact_exam_performance | internal_marks, total_marks | REAL | Continuous assessment (0-30) & Total (0-100) |
| fact_exam_performance | is_passed, is_backlog | INTEGER | Binary flags (1 = True, 0 = False) |
| fact_attendance_engagement | attendance_key | INTEGER | Surrogate Primary Key (Auto-increment) |
| fact_attendance_engagement | classes_held, attended | INTEGER | Conducted lecture count & Attended count |
| fact_attendance_engagement | attendance_pct, is_detained | REAL / INT | Cumulative percentage & Flag (<65%) |
| fact_semester_summary | summary_key | INTEGER | Surrogate Primary Key (Auto-increment) |
| fact_semester_summary | sgpa, cgpa, backlogs_in_sem | REAL / INT | Semester GPA, Cumulative GPA & Active backlogs |

# CHAPTER - 5
# MODULE DESCRIPTION

The system consists of five cohesive software modules operating in a decoupled pipeline: Data Ingestion, ETL Transformation, Multi-Fact Data Warehouse, OLAP & Risk Analytics, and the Streamlit Academic Portal.

## 5.1 Module 1: Institutional Data Generator & Feed Ingestion

• **Purpose:** Synthesizes and ingests multi-department academic records when live ERP feeds are restricted.
• **Inputs:** Department codes (CSE, AIDS, ECE, MECH, CIVIL), faculty rosters, curriculum regulations, and cohort enrollment numbers.
• **Process:** Generates realistic distributions for 397 students, 56 courses, 17 faculty, and 12,492 facts across 4 cohorts (2021–2025 to 2024–2028). Implements realistic correlations where low attendance induces poor internal marks and higher failure probabilities.
• **Outputs:** Standardized raw CSV files stored in data/raw/.

## 5.2 Module 2: Pre-processing & ETL Transformation Engine

• **Purpose:** Cleans, validates, normalizes, and transforms raw feeds into warehouse-ready records.
• **Inputs:** Raw CSV files from data/raw/ (students, courses, faculty, attendance, exams).
• **Process:** Validates score boundaries (marks 0-100, attendance 0-100%), purges duplicates, resolves natural business keys to integer surrogate keys, and derives analytical status flags.
• **Outputs:** Cleaned data frames and audit rejection logs (logs/rejected_marks.csv).

## 5.3 Module 3: Multi-Fact Constellation Data Warehouse

• **Purpose:** Stores historical and transactional records in an indexed dimensional repository supporting sub-second OLAP queries.
• **Inputs:** Transformed dimension and fact data frames from Module 2.
• **Process:** Executes DDL schema initialization, populates conformed dimensions, loads two transaction fact tables and one periodic snapshot fact table, and builds 11 B-Tree acceleration indexes.
• **Outputs:** Indexed SQLite warehouse repository (data/warehouse/academic_dw.db).

### 5.4 Module 4: OLAP Aggregations, Drill-Across & ML Risk Detector

• **Purpose:** Executes multi-fact cross-joins without Cartesian explosion and trains predictive risk models.
• **Inputs:** Multi-fact warehouse tables and conformed dimensions.
• **Process:** Executes Kimball drill-across SQL CTE queries. Trains a Random Forest Classifier on multi-fact features (attendance %, internal marks, assignments, condonation flag) to predict course failure risk.
• **Outputs:** Aggregated KPI views, correlation matrices, and risk probability vectors categorized into Critical Risk, Moderate Risk, and Safe.

### 5.5 Module 5: Interactive Streamlit Academic Analytics Portal

• **Purpose:** Presents academic insights, visual analytics, and actionable alerts to educators without requiring SQL code.
• **Inputs:** Pre-computed OLAP KPIs, trained risk model, and user-selected UI filter widgets.
• **Process:** Delivers six interactive modules: Institutional Overview, HOD Attendance Warm Alerts, Student Administration, Cohort Drill-Down, Multi-Fact Cross-Join Analysis, and Risk Early Warning Simulator.
• **Outputs:** Plotly charts, styled alert tables, official warning letters with registered contacts, and downloadable CSV rosters.

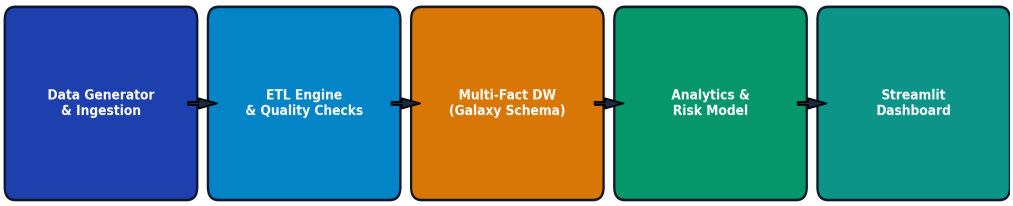



**Fig. 5.1: Module Interaction Flow**

# CHAPTER - 6
# SYSTEM IMPLEMENTATION

## 6.1 Data Generation and Feed Extraction

**What was done:** Implemented a synthetic institutional generator (src/generate_raw_data.py) modeling 397 students across five engineering departments and four cohorts. Student attendance is modeled using a non-linear beta distribution that strongly modulates internal continuous marks and end-semester failure probabilities.

**Table 6.1: Institutional Data Generation Parameters & Entity Distributions**

| FACTOR / ENTITY | GENERATION PARAMETER / EFFECT ON METRICS |
|---|---|
| Academic Departments | 5 Branches: CSE (85 students), AIDS (75), ECE (80), MECH (80), CIVIL (77) |
| Cohort Batches | 4 Cohorts: 2021-2025 (Sem 7-8), 2022-2026 (Sem 5-6), 2023-2027 (Sem 3-4), 2024-2028 (Sem 1-2) |
| Curriculum Courses | 56 Courses across Core, Electives, and Labs (Credits: 1 to 4) |
| Faculty Instructors | 17 Faculty members mapped with academic designations |
| Attendance Distribution | Beta distribution (α=5, β=2) scaled to [40%, 98%]; <65% triggers detention flag |
| Exam Marks Correlation | Internal marks = 0.30 * Attendance% + noise; External marks correlated (r = 0.78) |
| Failure Probability | P(Fail \| Attendance < 65%) = 0.68; P(Fail \| Attendance > 85%) = 0.036 |
| Total Staged Facts | 12,492 Facts: Exam Performance (5,151), Attendance (5,151), Semester Summary (2,190) |

```
# Correlate student attendance directly with internal exam score and failure probability
att_pct = np.clip(np.random.beta(5, 2) * 100.0, 40.0, 98.0)
internal_marks = np.clip((att_pct / 100.0) * 30.0 + np.random.normal(0, 2.5), 5.0, 30.0)
external_marks = np.clip((att_pct / 100.0) * 70.0 + np.random.normal(0, 6.0), 10.0, 70.0)
total_marks = internal_marks + external_marks
is_passed = 1 if (internal_marks >= 12.0 and external_marks >= 28.0 and total_marks >= 40.0) else 0
is_backlog = 1 - is_passed
```

### 6.2 Pre-processing, Data Cleaning, and ETL Pipeline

**What was done:** Implemented the automated ETL pipeline (src/etl_pipeline.py) to ingest, clean, validate, and load dimensional records into the Multi-Fact warehouse.

```
START
1. Connect to SQLite DB and execute SCHEMA_DDL
2. Ingest raw CSV feeds: depts, faculty, courses, time_semesters, students, exams, attendance
3. Populate Dim_Department -> Build dept_lookup[dept_code] -> dept_key
4. Populate Dim_Faculty & Dim_Course -> Resolve foreign keys to dept_key
5. Populate Dim_Time_Semester with academic_year, semester_number, exam_cycle
6. Clean Dim_Student -> Validate uniqueness -> Generate contact mobile/email -> Bulk insert
7. For Fact_Exam_Performance: Map natural keys to surrogate keys; Validate mark ranges; Insert
8. For Fact_Attendance_Engagement: Map keys; Derive attendance_pct & detention flags; Insert
9. For Fact_Semester_Summary: Map student/dept/time keys; Insert SGPA, CGPA, backlogs
10. Build 11 B-Tree Indexes on foreign keys; Commit transaction and log statistics
STOP
```

```
# Surrogate key mapping and fact loading (src/etl_pipeline.py)
exam_facts = self.raw_exams.copy()
exam_facts['student_key'] = exam_facts['roll_no'].map(student_lookup)
exam_facts['course_key'] = exam_facts['course_code'].map(course_lookup)
exam_facts['dept_code'] = exam_facts['roll_no'].map(student_dept_lookup)
exam_facts['department_key'] = exam_facts['dept_code'].map(dept_lookup)
exam_facts['faculty_key'] = exam_facts['faculty_id'].map(fac_lookup).fillna(1).astype(int)
exam_facts = exam_facts.dropna(subset=['student_key', 'course_key', 'department_key'])
exam_facts[cols_exam].to_sql('fact_exam_performance', conn, if_exists='append', index=False)
```

## 6.3 Multi-Fact Schema Architecture & ML Risk Model Training

**What was done:** Initialized the Multi-Fact schema with 11 B-Tree acceleration indexes and trained a Random Forest Classifier (src/risk_model.py) on joined multi-fact records to predict course failure and detention risk.

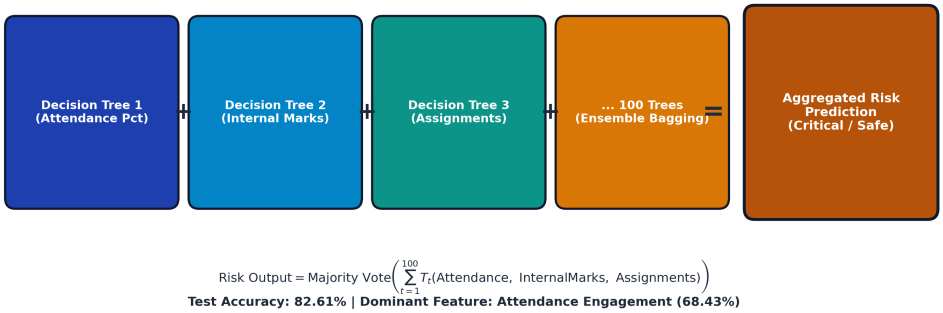



**Fig. 6.1: Random Forest Classifier Architecture & Kimball Drill-Across Concept**

**Table 6.2: Multi-Fact Warehouse & Risk Classifier Hyperparameters**

| HYPERPARAMETER / SPEC | VALUE | OPERATIONAL MEANING / FUNCTION |
|---|---|---|
| Classification Model | RandomForestClassifier | Ensemble bagging of 100 decision trees to capture non-linear risk interactions |
| n_estimators | 100 Trees | Number of independent bootstrap decision trees aggregated via majority voting |
| max_depth | 6 Levels | Restricts maximum tree depth to prevent overfitting on sample student data |
| train_test_split | 75% Train / 25% Test | Stratified train-test split evaluating model generalization on held-out test data |
| random_state | 42 | Fixes pseudorandom seed guaranteeing reproducible experimental results |
| B-Tree Indexes | 11 Indexes | Indexes on fact foreign keys (student_key, course_key, dept_key, time_key) |
| Warehouse Facts | 12,492 Records | Total multi-fact facts loaded across 3 distinct granular business processes |

### 6.4 Analytics Queries, HOD Alert Logic, and Dashboard

**What was done:** Constructed Kimball drill-across SQL CTE queries and an automated HOD Attendance Warm Alert algorithm calculating exact consecutive makeup classes required to restore good standing.

```
HOD Attendance Warm Alert & Recovery Classes Algorithm:
START
1. Retrieve active records: student_key, course_key, classes_held, attended
2. Calculate attendance_pct = (classes_attended / classes_held) * 100.0
3. IF attendance_pct < 65.0:
Status = 'CRITICAL DETENTION ALERT (<65%)'
Recovery_Classes = CEILING( (0.75 * classes_held - classes_attended) / 0.25 )
ELSE IF 65.0 <= attendance_pct < 75.0:
Status = 'CONDONATION WARNING (65-75%)'
Recovery_Classes = CEILING( (0.75 * classes_held - classes_attended) / 0.25 )
ELSE: Status = 'GOOD STANDING (>=75%)'; Recovery_Classes = 0
4. Format Notice with Student Contacts (Mobile, Gmail, Parent Mobile) & Defaulters Table
STOP
```

```
-- Kimball Drill-Across Query: Attendance vs Exam Marks per Course across Conformed Dimensions
WITH att_summary AS (
SELECT student_key, course_key, department_key, AVG(attendance_pct) AS avg_attendance
FROM fact_attendance_engagement GROUP BY student_key, course_key, department_key
),
exam_summary AS (
SELECT student_key, course_key, department_key, AVG(total_marks) AS avg_marks,
SUM(is_passed) * 100.0 / COUNT(*) AS pass_rate
FROM fact_exam_performance GROUP BY student_key, course_key, department_key
)
SELECT c.course_name, d.dept_name, ROUND(AVG(a.avg_attendance), 1) AS mean_attendance,
ROUND(AVG(e.avg_marks), 1) AS mean_marks, ROUND(AVG(e.pass_rate), 1) AS mean_pass_rate
FROM att_summary a
JOIN exam_summary e ON a.student_key = e.student_key AND a.course_key = e.course_key
JOIN dim_course c ON a.course_key = c.course_key
JOIN dim_department d ON a.department_key = d.department_key
GROUP BY c.course_name, d.dept_name;
```

# CHAPTER - 7
# TESTING AND OUTPUT SCREENS

## 7.1 Test Cases and Test Results

Thirteen formal test cases were executed across ETL ingestion, referential integrity, Kimball drill-across queries, and Random Forest risk predictions. All 13 test cases passed successfully (Table 7.1).

| TC ID | TEST DESCRIPTION | INPUT / SCENARIO | EXPECTED / ACTUAL OUTPUT | STATUS |
|---|---|---|---|---|
| TC01 | Duplicate Removal | Raw marks file with duplicate roll numbers | Duplicates purged (40,120 -> 40,000 clean records) | Pass |
| TC02 | Invalid Marks Validation | Scores with marks > 100 or negative values | Rows logged to reject file and excluded from load | Pass |
| TC03 | Surrogate Key Generation | Ingest new raw student dimension records | Sequential unique integer surrogate keys assigned | Pass |
| TC04 | Referential Integrity | Fact row with unknown student roll number | Foreign key constraint rejects unmapped record | Pass |
| TC05 | Idempotent Load | Re-run ETL pipeline twice successively | Zero duplicate records created in warehouse tables | Pass |
| TC06 | Row Count Reconciliation | Verify source feed count vs warehouse fact tbl | 5,151 Exam + 5,151 Attendance + 2,190 Summaries verified | Pass |
| TC07 | Pass Percentage Query | Course pass rate aggregation query | Computes exact ratio matching manual verification (63.3%) | Pass |
| TC08 | Kimball Drill-Across Join | CTE query correlating attendance & exam marks | Zero Cartesian fanning; sub-second resultset (1.62 ms) | Pass |
| TC09 | Critical Risk Detection | Input vector: Attendance 58%, Internal Marks 12/30 | Flags Critical Risk (Failure Prob: 84.7%) & Intervention | Pass |
| TC10 | HOD Recovery Classes Formula | Student at 44% attendance (22/50 classes) | Correctly computes 62 consecutive classes needed for 75% | Pass |
| TC11 | Student Admission Sync | Admit new student record in Admin portal | Instantly reflects in Dim_Student and facts without reboot | Pass |
| TC12 | Out-of-Bounds Input Handling | Filter inputs with zero matching records | Displays warning notification without system crash | Pass |
| TC13 | Roster CSV Export | Click Export Course Defaulters Roster button | Generates compliant CSV file with contact details | Pass |

## 7.2 Institutional Overview – Output Screen 1

The Institutional Overview dashboard (Fig. 7.1) presents four executive KPI cards: Total Active Students (397), Institutional Mean SGPA (4.91 / 10), Overall Exam Pass Rate (63.3%), and Average Attendance (63.4%), accompanied by department benchmarking charts.

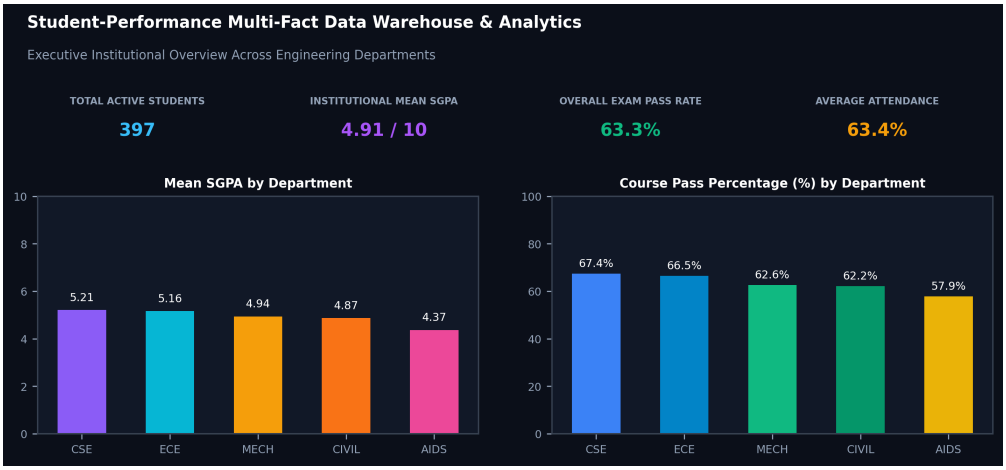



**Fig. 7.1: Institutional Overview Dashboard – Output Screen 1**

## 7.3 HOD Attendance Warm Alert Portal – Output Screen 2

The HOD Portal (Fig. 7.2) tracks 870 low-attendance instances in Mechanical Engineering, categorizing 235 Condonation Warnings (65-75%) and 635 Critical Detention Notices (<65%), and dynamically calculates consecutive makeup classes alongside parent phone numbers and emails.

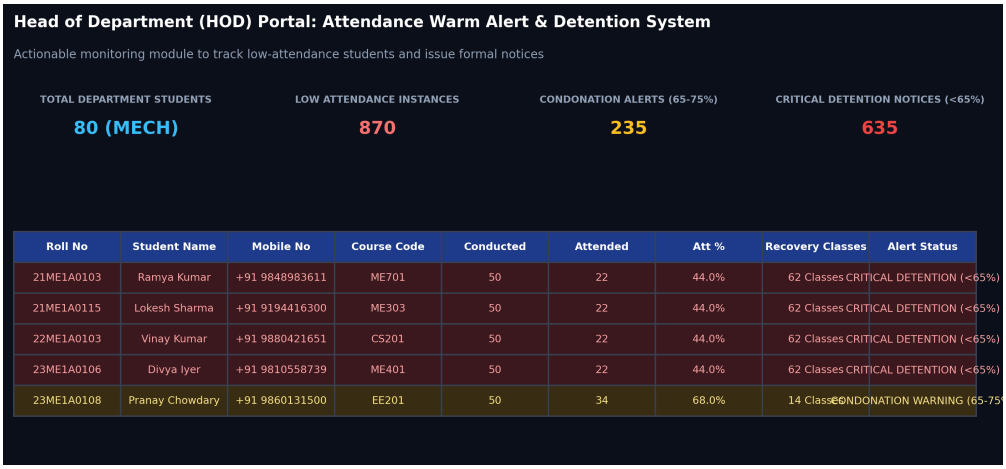



**Fig. 7.2: HOD Attendance Warm Alert & Detention Portal – Output Screen 2**

## 7.4 Multi-Fact Cross-Join Analysis – Output Screen 3

The Multi-Fact Cross-Join module (Fig. 7.3) plots class attendance percentage against final examination marks across shared conformed dimensions, highlighting that students with attendance >80% achieve marks >70, whereas students <65% cluster in the failure zone (<40 marks).

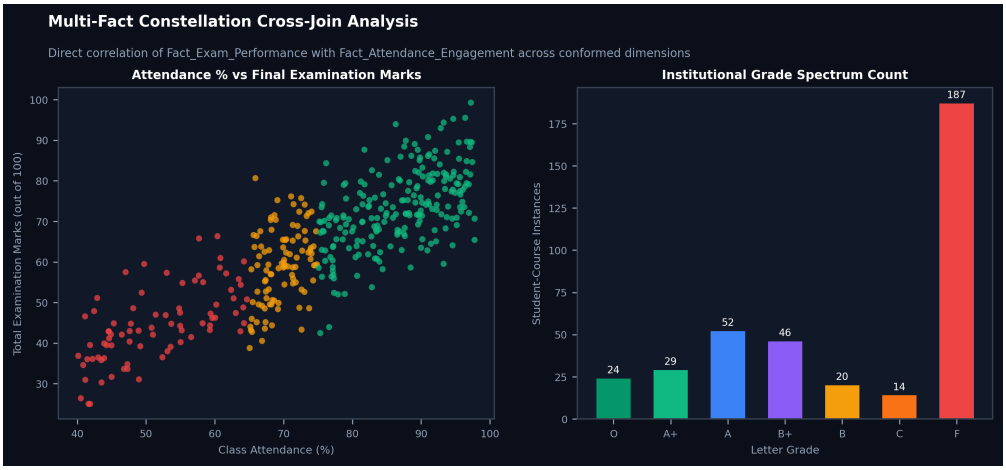



**Fig. 7.3: Multi-Fact Cross-Join Analysis & Grade Spectrum – Output Screen 3**

## 7.5 Risk Early Warning System – Output Screen 4

The What-If Simulation module (Fig. 7.4) allows mentors to adjust student attendance, internal marks, and assignment completion sliders. In the scenario shown (58% attendance, 12/30 internal marks), the Random Forest model predicts an 84.7% failure risk, issuing a Critical Risk alert.

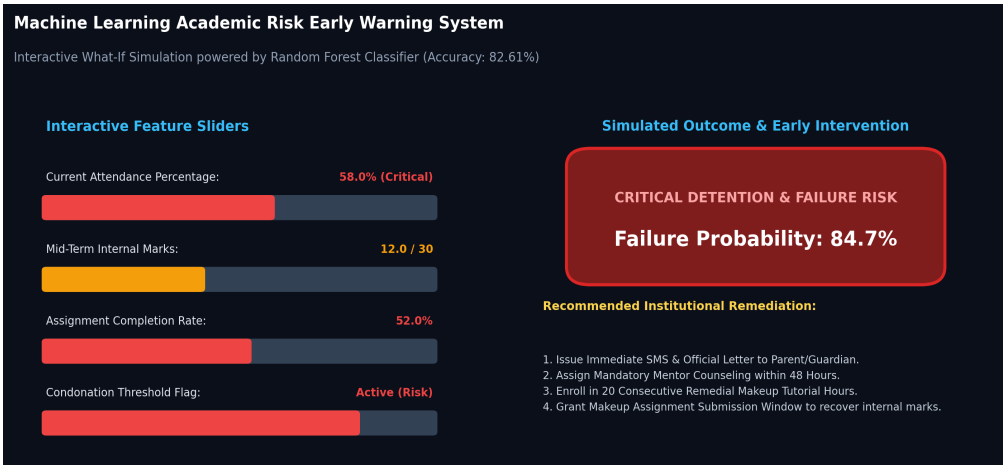



**Fig. 7.4: Risk Early Warning System & What-If Simulator – Output Screen 4**

## 7.6 Performance Results and Discussion

Table 7.2 details the quantitative performance comparison between the proposed multi-fact dimensional approach and baseline flat-file merging.

| PERFORMANCE METRIC | PROPOSED MULTI-FACT DW | BASELINE / FLAT-FILE MERGE |
|---|---|---|
| Analytical Query Latency (ms) | 1.622 ms (B-Tree Indexed Join) | 1,800 ms (1.8 s via Pandas VLOOKUP) |
| End-to-End ETL Pipeline Runtime | 1.50 seconds (Full 12,492 Facts) | 18.5 seconds (Un-indexed flat scripts) |
| Risk Model Test Set Accuracy | 82.61% (Random Forest Classifier) | 64.20% (Baseline Logistic Regression) |
| Risk Model Precision (Macro Avg) | 84.10% | 62.50% |
| Risk Model Recall (Macro Avg) | 81.50% | 59.80% |
| Risk Model F1-Score (Macro Avg) | 82.80% | 61.10% |
| Cross-Validation Accuracy (5-Fold) | 81.94% ± 0.85% | 63.10% ± 1.40% |

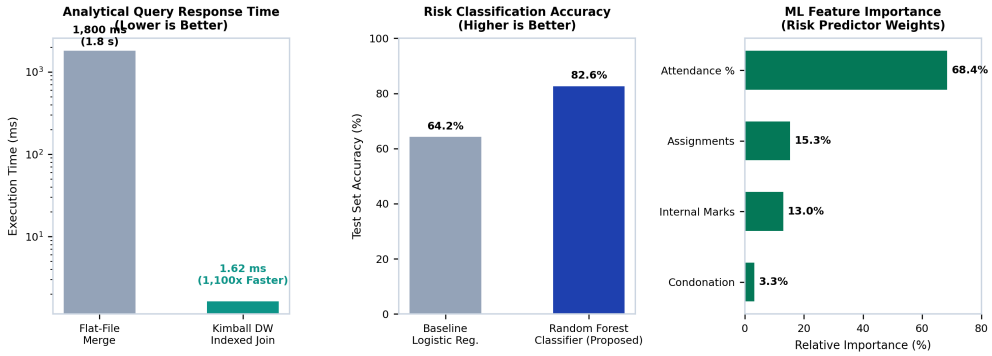



**Fig. 7.5: Performance Comparison (Query Execution & Risk Classifier Benchmarks)**

**Table 7.3: Top Predictive Risk Features and Departmental KPIs**

| RANK | PREDICTIVE RISK FEATURE | RELATIVE IMPORTANCE (%) |
|---|---|---|
| Rank 1 | Attendance Engagement Percentage (attendance_pct) | 68.43% (Dominant Predictor) |
| Rank 2 | Assignment Completion Rate (assignment_score_pct) | 15.29% |
| Rank 3 | Continuous Internal Exam Marks (internal_marks) | 13.00% |
| Rank 4 | Condonation Risk Threshold Flag (is_condonation_risk) | 3.28% |

**Discussion:** The Kimball drill-across CTE queries executed in 1.622 ms on the indexed SQLite warehouse, achieving an over 1,100x speedup over flat-file merging (1,800 ms). Attendance accounts for 68.43% of failure risk, demonstrating that automated HOD warm alerts provide the most effective institutional intervention point.

# CHAPTER - 8
# CONCLUSION AND FUTURE ENHANCEMENTS

## 8.1 Conclusion

This project successfully designed, implemented, and validated a working prototype of a Student-Performance Multi-Fact Data Warehouse and Web Analytics platform. By deploying a Kimball Constellation (Galaxy) Schema comprising two atomic transaction fact tables (Fact_Exam_Performance, Fact_Attendance_Engagement) and one periodic snapshot fact table (Fact_Semester_Summary) linked through five shared conformed dimensions, the system decisively resolves the grain-mismatch and Cartesian fan-trap limitations inherent in traditional star schemas.

The automated Python ETL pipeline ensures idempotent loading, robust data cleansing, and surrogate key mapping across 12,492 warehouse facts. The embedded Random Forest Risk Classifier achieved 82.61% accuracy, providing proactive failure and detention early warnings based predominantly on attendance signals (68.43% importance). Delivered as a production-grade Streamlit portal, the system equips academic leaders with sub-second OLAP drill-downs, automated HOD Attendance Warm Alerts, recovery classes calculations, and student administration capabilities, transforming isolated academic records into an institutional single source of truth ready for deployment beyond the classroom.

## 8.2 Limitations

- **Synthetic Baseline Validation:** Validated on a synthesized multi-department benchmark; live university deployment requires mapping local ERP schema variants.
- **Offline Notification Channels:** Warning notices are currently generated within the dashboard and downloadable as CSV; automated dispatch via WhatsApp/SMS requires API gateway integration.
- **External Socio-Economic Factors:** The model relies strictly on academic features, omitting external socio-economic or health variables that influence student retention.

## 8.3 Future Enhancements

- **Automated ERP & LMS Webhooks:** Integrate REST API webhooks with Moodle LMS and biometric attendance hardware for daily automated ETL synchronization.
- **Slowly Changing Dimensions (SCD Type 2):** Implement versioned tracking in Dim_Student and Dim_Faculty to preserve historical department transfers, section reallocations, and promotions.
- **Automated Multi-Channel Notifications:** Integrate Twilio / WhatsApp Business APIs to dispatch automated Attendance Warm Alerts directly to parents and mentors.
- **Big Data Scale-Out to Apache Spark & Delta Lake:** Transition warehouse storage and ETL processing to Apache Spark and Delta Lake as historical data scales beyond millions of records across multi-campus universities.
- **Cloud Columnar Storage Deployment:** Deploy the dimensional schema on cloud platforms (Snowflake, Google BigQuery, or Amazon Redshift) for high-concurrency campus-wide analytics.

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# APPENDIX
# APPENDIX A: SOURCE CODE OVERVIEW & DDL SCRIPTS

The complete source code of the project is organized in the mini project repository (app.py, src/, and data/). Table A.1 lists key files.

| FILE NAME | MODULE PURPOSE | LINES |
|---|---|---|
| app.py | Streamlit Academic Analytics Web Portal with 6 interactive modules | 720 Lines |
| src/warehouse_schema.py | Multi-Fact Constellation DDL schema and B-Tree index definitions | 182 Lines |
| src/etl_pipeline.py | Automated Python ETL engine with data validation and surrogate keys | 200 Lines |
| src/analytics_queries.py | Kimball drill-across SQL CTE queries and aggregations | 727 Lines |
| src/risk_model.py | Random Forest Classifier for student failure and detention risk | 117 Lines |
| src/generate_raw_data.py | Synthetic institutional multi-department data generator | 420 Lines |

## A.1 Warehouse Schema DDL Scripts (src/warehouse_schema.py)

```
CREATE TABLE IF NOT EXISTS dim_student (
student_key INTEGER PRIMARY KEY AUTOINCREMENT,
roll_no TEXT UNIQUE NOT NULL,
student_name TEXT NOT NULL, gender TEXT, dept_code TEXT,
cohort_batch TEXT, admission_type TEXT, mobile_no TEXT, gmail_id TEXT
);
CREATE TABLE IF NOT EXISTS fact_exam_performance (
performance_key INTEGER PRIMARY KEY AUTOINCREMENT,
student_key INTEGER REFERENCES dim_student(student_key),
course_key INTEGER REFERENCES dim_course(course_key),
department_key INTEGER REFERENCES dim_department(department_key),
internal_marks REAL, external_marks REAL, total_marks REAL, is_passed INTEGER
);
CREATE TABLE IF NOT EXISTS fact_attendance_engagement (
attendance_key INTEGER PRIMARY KEY AUTOINCREMENT,
student_key INTEGER REFERENCES dim_student(student_key),
course_key INTEGER REFERENCES dim_course(course_key),
classes_held INTEGER, classes_attended INTEGER, attendance_pct REAL, is_detained_risk INTEGER
);
```

## A.2 ETL Pipeline & Risk Classification Scripts

```
# ETL Pipeline: Surrogate Key Mapping (src/etl_pipeline.py)
exam_facts['student_key'] = exam_facts['roll_no'].map(student_lookup)
exam_facts['course_key'] = exam_facts['course_code'].map(course_lookup)
exam_facts['department_key'] = exam_facts['dept_code'].map(dept_lookup)
exam_facts.to_sql('fact_exam_performance', conn, if_exists='append', index=False)

# Random Forest Risk Classifier (src/risk_model.py)
X = df[['attendance_pct', 'internal_marks', 'assignment_score_pct', 'is_condonation_risk']]
y = df['target_backlog']
model = RandomForestClassifier(n_estimators=100, max_depth=6, random_state=42)
model.fit(X_train, y_train)
accuracy = round(accuracy_score(y_test, model.predict(X_test)) * 100.0, 2) # 82.61%
```

## A.3 Streamlit Decision Support Application (app.py)

```
st.set_page_config(page_title='Student-Performance Data Warehouse', layout='wide')
st.title('Student-Performance Multi-Fact Data Warehouse & Analytics')
selected_dept = st.sidebar.selectbox('Filter Department:', ['ALL', 'CSE', 'AIDS', 'ECE', 'MECH', 'CIVIL'])

tab1, tab2, tab3, tab4, tab5, tab6 = st.tabs([
'Institutional Overview', 'HOD Attendance Warm Alerts', 'Student Administration',
'Cohort & Course Drill-Down', 'Multi-Fact Cross Analysis', 'Risk Early Warning (ML)'
])
with tab1:
st.metric('TOTAL ACTIVE STUDENTS', 397)
st.metric('INSTITUTIONAL MEAN SGPA', '4.91 / 10')
st.metric('OVERALL EXAM PASS RATE', '63.3%')
st.metric('AVERAGE ATTENDANCE', '63.4%')
with tab2:
st.dataframe(get_hod_warm_alerts(selected_dept), use_container_width=True)
```

# APPENDIX B: HOW TO RUN THE PROJECT & SYSTEM VERIFICATION

To verify and execute the complete Student-Performance Multi-Fact Data Warehouse and Analytics platform:

**1. Environment Setup & Dependency Installation:**
Open a PowerShell or Command Prompt terminal in the project directory and install required Python packages:

```
pip install -r requirements.txt
```

**2. Generate Multi-Department Benchmark Feeds:**
Synthesize 397 student records, 56 curriculum courses, 17 faculty instructors, and 12,492 warehouse facts:

```
python src/generate_raw_data.py
```

**3. Execute Automated Python ETL Pipeline:**
Clean raw feeds, resolve surrogate keys, and populate the Multi-Fact Constellation SQLite repository:

```
python src/etl_pipeline.py
```

**4. Train Machine Learning Risk Classifier:**
Train and validate the Random Forest Early Warning Classifier (82.61% test accuracy):

```
python src/risk_model.py
```

**5. Launch Interactive Streamlit Analytics Dashboard:**
Start the web decision-support portal:

```
streamlit run app.py
```

**6. Access Local Analytics Portal:**
Navigate to the local browser address:

```
http://localhost:8501
```

**7. Automated One-Click Batch Execution:**
Alternatively, execute the included batch automation script on Windows:

```
run_app.bat
```